

Sreya Sree Siloju

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Results-driven Data analyst and Engineer with expertise in designing, developing, and optimizing data pipelines, **ETL processes**, and cloud-based data solutions. Proficient in **SQL, Python, and Big Data technologies** for building scalable and efficient data architectures. Experienced in **data modeling, warehouse management, and real-time data processing** using **Azure, AWS, and Google Cloud**. Adept at collaborating with cross-functional teams to ensure **data integrity, security, and governance** while delivering actionable insights for business intelligence.

SKILLS

- **OS:** Windows
- **Programming:** SQL, Python, R, Excel, Scala, Java, React
- **ETL & Data Integration Tools:** Informatica PowerCenter, Azure Data Factory, Snowflake, Apache Airflow, Talend, SSIS, SSMS, Alteryx, Unity Catalog
- **Data Modeling Tools:** Erwin, Power Designer, Star & Snowflake Schema, Kimball Methodology
- **Databases:** SQL (PostgreSQL, MySQL, SQL Server, Oracle, Teradata, MS Access), NoSQL (MongoDB, Cassandra), OLAP, OLTP
- **Version control and Others:** Git, GitHub, GitLab, Jira, MS Project, MS Office, Docker, Kubernetes
- **Big Data:** Azure, AWS (Glue, Lambda, S3, EC2, Redshift, RDS), Google Cloud (BigQuery, DataFlow), Apache Spark, Hadoop
- **Data Analytics & Modeling:** Statistical Analysis, Predictive analysis, Clustering, Machine Learning, Customer Segmentation, Structures and Unstructured file handling (Flatfile, JSON, XML, Parquet, CSV, PDF)
- **Data Visualization:** Power BI, Tableau, MS Excel, Looker, Google Data Studio, PowerApps
- **Methodologies:** SDLC, JAD Sessions, Agile (Scrum, Kanban), DAX, Data Governance, Data Quality Management

PROFESSIONAL CERTICATIONS

Microsoft: Azure Data Engineer Associate (ETL pipelines, data lakes, Synapse Analytics)

Coursera: IBM Data Science Professional Certificate (SQL, Python, data wrangling, and visualization)

EXPERIENCE

Guest Researcher | UMBC | Baltimore, MD

Feb 2025 - Present

- Built a **Generative AI assistant** that achieved **90% accuracy** in converting natural language to optimized SQL queries.
- Developed cross-platform query generation for **PostgreSQL, MySQL, and Snowflake**, boosting SQL efficiency by **40%**.
- Reduced SQL execution time by **60%** through indexing, optimized joins, and execution plan analysis.
- Designed a **Streamlit-based UI** for real-time SQL generation and explanations, increasing adoption by **70%**.
- Integrated **AI-driven query insights** with **ELK** and **Splunk**, enhancing user understanding by **50%**.
- Implemented adaptive feedback loops, improving model accuracy by **20%** over iterations.
- Deployed scalable solutions using **Docker, AWS, and GCP**, maintaining **99.9% uptime**.

Data Analyst | M&T Bank | Baltimore, MD

Jan 2024 – Dec 2024

- Automated 30+ ETL/ELT data pipelines across **Azure, AWS, and GCP**, reducing data latency by **45%** and improving reliability through orchestration with **Apache Airflow** and **Talend**.
- Improved SQL query performance by **40%** using advanced tuning techniques (indexing, partitioning) on **PostgreSQL, SQL Server, and Oracle** databases.
- Architected and managed **cloud-native data warehouses** (Azure Synapse, AWS Redshift, BigQuery) to support real-time analytics for over **10 million rows** of transactional data daily.
- Designed and deployed data lake strategies using **Azure Data Factory, AWS S3, and Google Cloud Storage**, improving data ingestion efficiency by **35%**.
- Engineered dimensional data models (Star, Snowflake schemas) using **Kimball methodology**, improving analytical query performance by **60%**.
- Developed 50+ dynamic dashboards using **Power BI, Tableau, and Looker**, enabling leadership to access insights with a **30% boost** in decision-making speed.
- Implemented ML-based data validation and anomaly detection scripts in **Python**, improving data quality assurance accuracy by **25%**.

- Led **data governance initiatives**, establishing ERDs, data dictionaries, and metadata standards, enhancing data consistency across 15+ projects.
- Administered **Power BI Service** and **Tableau Server**, automating refresh cycles and user permissions for 200+ stakeholders.

Data Analyst | ZF India Private Ltd | Hyderabad, India

July 2020 – Jan 2023

- Built and maintained 40+ scalable ETL pipelines using **Azure Data Factory**, **SSIS**, and **Informatica PowerCenter**, reducing manual data processing by **70%**.
- Enhanced stored procedure performance by **35%** across **Teradata**, **SQL Server**, and **PostgreSQL** by optimizing queries and execution plans.
- Designed and implemented data models using **Kimball principles** across **3 enterprise warehouses**, improving data accessibility by **50%**.
- Orchestrated batch and streaming data workflows with **Apache Airflow**, increasing pipeline reliability and scheduling success rate by **95%**.
- Created 20+ interactive dashboards in **Power BI** and **Tableau**, integrating storytelling features (drill-downs, bookmarks, filters) that improved user engagement by **30%**.
- Developed custom Python and R scripts to automate data cleaning, transformation, and anomaly detection, improving data accuracy by **40%**.
- Spearheaded data governance policy rollout, increasing compliance and reducing duplication errors by **20%**.
- Integrated big data tools including **Databricks**, **Azure Synapse**, and **AWS Redshift** to process over **2 TB** of data monthly for business reporting.

Data Analyst Intern | iCIMS | Hyderabad, India

September 2019 – June 2020

- Developed and deployed ETL pipelines using **Informatica PowerCenter**, **Apache Airflow**, and **SSIS**, improving integration efficiency by **25%**.
- Improved SQL query efficiency by **20%** through optimized logic and indexing across **SQL Server**, **Oracle**, and **PostgreSQL**.
- Automated reporting workflows with Python and R, cutting manual reporting time by **30 hours/month**.
- Built statistical models to forecast incident trends, increasing workforce planning accuracy by **10%**.
- Enabled data visualizations with **Power BI**, **Tableau**, and **Cognos**, leading to **30% higher adoption** of analytical dashboards.
- Supported cloud-based analytics with **AWS Redshift**, **BigQuery**, and **Azure Synapse**, scaling processing for 5+ cross-functional teams.

EDUCATION

Master of Science in Data Science | University of Maryland, Baltimore County, Baltimore, MD

Jan 2023 – Dec 2024

- **GPA:** 3.91
- Coursework: Data modelling, data Warehousing, Data administration

Bachelor of Engineering in Electronics and Communication Engineering | Osmania University, Hyderabad, India

Aug 2016 – Apr 2020

- **GPA:** 3.36
- Coursework: Programming (Python, C), Data Structures, RTOS, AI

AWARDS AND RECOGNITIONS

- **STAR Award** – Recognized as part of the best-performing team in the division for delivering outstanding outcomes.
- **SPOT Award** – Received for exceptional contributions to the organization, leading to successful achievement of key objectives.